# Beyond Raw Transcripts: Structured Persona Extraction for LLM-Based Digital Twins

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## Abstract

LLM-based "digital twins" aim to simulate how an individual would behave in new environments or respond to novel questions, given some representation of that individual's prior responses. A common approach constructs this representation from survey transcripts or summaries derived from them, and evaluates performance by predicting holdout responses. Prior work shows that compressing long transcripts into shorter LLM-generated summaries does not significantly reduce predictive accuracy, suggesting that information volume is not the primary bottleneck.

In this work, we argue that the key limitation is instead structural: how persona information is organized before being provided to the simulator model. We study this by comparing unstructured summaries with structured persona representations. First, we introduce a hand-crafted schema (BDE: Background, Decision procedure, Evaluation), grounded in consumer-behavior theory, and show that it improves predictive accuracy over raw transcripts by $+1.91$ percentage points on a homogeneous benchmark (Twin-2K-500), with similar gains on `gpt-5.4-mini` and `Qwen3-8B` as robustness checks. However, this fixed structure does not generalize across more heterogeneous tasks, where performance is statistically indistinguishable from the raw transcript baseline.

To address this limitation, we propose an automatic structure-discovery pipeline in which an LLM iteratively proposes and refines task-specific persona structures and extraction prompts. On a benchmark of 13 diverse sub-studies, this approach restores performance, improving mean accuracy by $+1.91$ percentage points over the raw transcript baseline and eliminating significant losses observed with the fixed schema.

Overall, our results suggest that the main constraint in LLM-based digital twins is not how much information is provided, but how it is structured—and that the optimal structure depends on the task.

## 1 Introduction

### 1.1 The persona-structure gap

LLM-based digital twins aim to simulate how an individual would behave or respond in new settings, given a representation of their prior data. A common instantiation elicits a long survey from each respondent, passes the resulting transcript to a language model, and asks it to predict holdout responses as that respondent would; this is the setting we study. The Twin-2K-500 dataset Toubia et al. [2025b] pairs 500 input questions with 88 holdout questions across 17 distinct prediction tasks for over two thousand respondents, offering substantially broader per-individual coverage than prior digital-twin benchmarks. A follow-up Mega-Study Toubia et al. [2025a]—19 pre-registered sub-studies spanning diverse decision domains—reports that a 13K-character LLM-generated persona summary performs

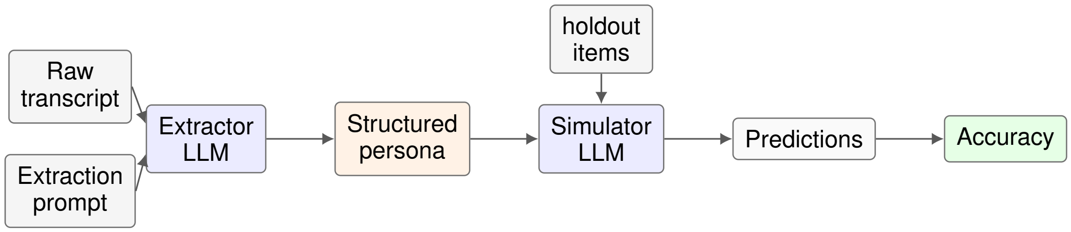


Figure 1: Pipeline: from raw transcript to accuracy via the structured persona.

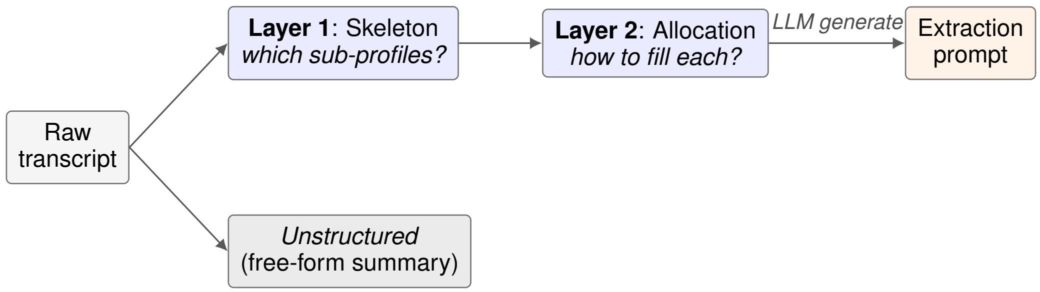


Figure 2: The two-layer view of persona structure: Layer 1 (skeleton) and Layer 2 (allocation), informed by the downstream queries, generate the extraction prompt that produces a *structured* persona; an *unstructured* persona is a free-form summary that bypasses both layers.

comparably to the 128K-character raw transcript(same as 500 input questions in twins-2k-500 Toubia et al. [2025b]) on prediction accuracy. This robustness to compression implies that information volume is not the binding constraint on digital-twin accuracy. What remains open is how that information should be organized before it reaches the simulator.

In this paper, we focus on this question and explore how to structure the persona information to improve the accuracy of persona based simulators. In our approach, for each respondent, an extractor LLM takes the raw transcript and an extraction prompt as input and produces a *structured persona*. The simulator conditions on this structured persona at test time, and its predictions on holdout items are scored to assess accuracy (Figure 1). The core design choice in an extraction prompt is its *structure*, which decomposes into two layers (Figure 2).

**Layer 1 (Skeleton).** This layer decides which sub-profiles, if any, to extract given the downstream decision problem or prediction queries that the extracted personas will face. For example, the Background–Decision procedure–Evaluation (BDE) structure uses three sub-profiles representing identity, reasoning, and preference.

**Layer 2 (Allocation).** Given the Layer 1 skeleton, this layer decides how raw transcript evidence is mapped into each sub-profile. For example, BDE's Background sub-profile is filled using demographics, Big Five personality, values, and political-belief items from the corresponding transcript sections.

Given a structure, the extraction prompt instructs the extractor LLM to produce a persona partitioned according to the Layer 1 skeleton, with each sub-profile filled using the transcript evidence specified by Layer 2. Sections 3.2 and 3.3 develop two concrete instantiations of this framework—BDE and an auto-discovered structure—which we use in the two main experiments. We call a persona *unstructured* if it is generated as a free-form document directly from the raw transcript, without using the downstream prediction questions to impose an explicit skeleton or allocation rule.

### 1.2 Contributions

**A fixed structure helps on a homogeneous task family but may not generalize to heterogeneous ones (§4).** On Twin-2K-500's 17 prediction tasks, the Background–Decision procedure–Evaluation (BDE) structure, which separates persona information into identity, reasoning, and preference sub-profiles, beats the unstructured baseline by $+2.49$pp and the raw transcript by $+1.91$pp on overall accuracy (full breakdown by metric and contrast in Table 2); the result reproduces on `gpt-5.4-mini` and `Qwen3-8B` (Appendix C). Applied to the 19 heterogeneous Mega-Study sub-studies, however, the same BDE essentially ties raw on the 19-study aggregate (Table 3).

**Per-task auto-discovery generalizes across heterogeneous tasks (§5).** A reflective refinement loop searches Layers 1 and 2 per sub-study. Aggregate accuracy lifts $+1.91$pp over raw, $+2.22$pp over hand-crafted BDE, and $+1.40$pp over the unstructured baseline (Table 4); the result reproduces on `gpt-5.4-mini` and `Qwen3-8B` (Appendix C).

**Released artifacts.** We will release the BDE extraction prompts, the auto-discovery pipeline implementation, and the 19 auto-discovered per-sub-study structures upon acceptance, as drop-in baselines for future persona-side prompt design.

## 2 Related Work

**LLM-based digital twins.** Work on LLMs as human surrogates ranges from aggregate market-respondent simulation Brand et al. [2023] to individual-level generative-agent populations Park et al. [2024]. Twin-2K-500 Toubia et al. [2025b] introduces the per-respondent prediction setting we build on: each LLM "twin" is conditioned on a rich transcript of one respondent's prior survey answers and asked to predict that respondent's holdout responses. The follow-up Mega-Study Toubia et al. [2025a] runs 19 pre-registered sub-studies—spanning domains as varied as misinformation sharing, luxury consumption, hiring algorithms, redistribution preferences, and narrative belief—and finds that twins' overall accuracy approaches the human test-retest ceiling but that adding richer persona content primarily improves between-participant correlation rather than individual-level accuracy. This study doesn't answer *how* persona information is organized and leaves it as an open question.

**Prompt design and reflective prompt evolution.** The form of an LLM's input—not just its content—shapes its generated behavior. Brucks and Toubia Brucks and Toubia [2025] document that prompt architecture creates large, model-consistent artifacts in LLM responses, and Gui and Toubia Gui and Toubia [2023] frame LLM-based simulation as a causal-inference problem in which under-specified prompts induce omitted-variable bias and over-specified prompts induce focalism. A recent thread takes this further by treating the prompt itself as a learnable artifact, optimized via natural-language reflection on rollouts rather than gradient signal: GEPA Agrawal et al. [2026] evolves a module's prompt against holdout reward, DSPy Khattab et al. [2024] treats prompts as code-like optimizable objects, and Voyager Wang et al. [2023] grows a per-task skill library reused across episodes. Our auto-discovery pipeline follows the same broad principle as GEPA: it uses natural-language reflection on rollout feedback to evolve prompts over iterations. In our setting, this principle is applied to an upstream persona-extraction problem. The evolved prompt determines which categories of behavioral evidence the persona contains and how they are organized, while the downstream simulator is kept fixed. Our goal is to use reflective prompt evolution as a tool for discovering sub-study-specific representation structures for digital-twin simulation.

**Behavioral-science foundations of the BDE Structure.** For the consumer-behavioral outcomes Twin-2K-500 evaluates, an LLM conditioned on a persona may need to recover three behaviorally distinct signals: identity (*who* the respondent is), reasoning procedure (*how* they decide), and preferences (*what* they want). Each axis is independently grounded in consumer-behavior theory: identity in the Engel–Kollat–Blackwell model Engel et al. [1968] and the Theory of Planned Behavior Ajzen [1991]; reasoning in Behavioral Reasoning Theory Westaby [2005] and the adaptive-decision-maker framework Payne et al. [1993]; preferences in Falk et al. Falk et al. [2018] and the constructive-choice account of Bettman, Luce and Payne Bettman et al. [1998]. A free-form summary (i.e. an unstructured persona) leaves these axes entangled; the BDE structured persona separates them once, upstream of the simulator.

**Persona axes in LLM internals.** The multi-axis structure may also align with how LLMs process personas: Anthropic's recent persona-axis research Lu et al. [2026] finds that LLM behavior is partly governed by selection among internalized persona axes, so a multi-axis structured persona may activate the matching combination rather than describing a person to a blank-slate predictor.

Read through this lens, BDE is a fixed recipe that exposes three behavioral axes simultaneously, and the auto-discovery pipeline is a per-task search for the axis combination that best activates correct behavior on the task's evidence.

# 3 Prerequisites and Experiment Design

## 3.1 Datasets and Evaluation

**Twin-2K-500.** We use Twin-2K-500 Toubia et al. [2025b] as the source of all persona inputs and holdout tasks: 500 input questions per respondent and an 88-question holdout evaluation block across 17 prediction tasks. The canonical baseline supplies the raw question–answer transcript verbatim as the persona-context component of the simulator prompt. We report three per-persona accuracy metrics, averaged over the $n=50$ persona panel: *overall* (the 17 prediction tasks), *cognitive-bias* (the holdout heuristics-and-biases items), and *pricing* (the 40-item willingness-to-pay block). The per-item scoring rule and the bias / pricing item lists are in Appendix A; examples for both metrics are reproduced in Appendix G.

**Mega-Study.** The Mega-Study Toubia et al. [2025a] runs 19 pre-registered sub-studies spanning misinformation sharing, luxury consumption, hiring algorithms, redistribution preferences, and narrative belief, which makes them a natural test of generalization for a single hand-crafted persona structure. Each sub-study draws its own panel of $n=50$ personas from the Twin-2K-500 pool and is scored under the Mega-Study's range-normalized accuracy. We compute overall accuracy across sub-studies using the row convention of Toubia et al. [2025a]. Full details are in Appendix A.

## 3.2 BDE structure

We define the BDE structure as a three-part persona representation. Layer 1 partitions the persona into three sub-profiles: background (bg), decision procedure (dp), and evaluation profile (ep). These sub-profiles correspond to the behaviorally distinct axes motivated in §2: identity, reasoning procedure, and preference. Layer 2 specifies how each sub-profile is filled from the respondent's input-side Twin-2K-500 question–answer transcript and the text of the downstream prediction questions, without access to the respondent's holdout answers. Appendix B gives a worked example of the resulting three-part representation for an anonymized respondent. We next detail the Layer 2 content of each sub-profile.

**Background profile (bg).** *The bg profile encodes who the respondent is, what they know, and what they believe.* It is filled using demographics and Big Five personality, the 24-item values scale as directly reported, political and policy beliefs, factually tested knowledge such as financial literacy and basic probability, and three self-report reliability indicators: acquiescence, social desirability, and the gap between stated and behavioral preferences from the input-side raw questions.

**Decision-procedure profile (dp).** *The dp profile encodes how the respondent reasons under uncertainty.* It is filled using evidence about reasoning style, effort regulation, closure tolerance, and decision strategy: need for cognition and cognitive-test error patterns, need for closure, the maximizer scale, the intertemporal-choice block read as a cognitive-style signal, the behavioral-vs.-stated preference gap, and per-bias susceptibility predictions with explicit confidence levels—high for biases empirically tied to cognitive ability, low otherwise from the input-side raw questions.

**Evaluation profile (ep).** *The ep profile encodes what the respondent wants and would pay for.* It is filled using evidence about budget constraints, value priorities, consumer orientation, risk and time preferences, and social preferences: the budget constraint, the values hierarchy read as relative structure after correcting for acquiescence, minimalism and consumer orientation, lottery-based risk preferences, intertemporal patience read as a preference, context-dependent social preferences toward strangers, family, and abstract-cause recipients, and the 40-item willingness-to-pay block from the input-side raw questions.

In experiments, the extraction prompt is applied once per persona to produce the BDE structured persona, which is then supplied to the simulator LLM in place of the raw transcript and scored against holdout responses.

### 3.3 Auto-discovered structure

For each Mega-Study sub-study, an *auto-discovered* structure is produced by an LLM-driven prompt-evolution procedure. The procedure first generates an initial round-0 structure, then applies four refinement rounds, and finally selects one of the five candidate rounds for the locked headline evaluation defined in §5.1.

1. **Initial structure.** A round-0 structure is LLM-generated once at the beginning: Layer 1 is derived from the summary of each sub-study's pre-registered *constructs*—the theorized behavioral concepts the sub-study targets (e.g., fairness perception, trust in algorithms), as defined in Toubia et al. [2025a]—and Layer 2 is derived from both the Layer 1 skeleton and the raw question.
2. **Round structure.** At each iteration, a meta-extractor LLM updates the structure at two levels: it adds or removes sub-profiles at Layer 1, and revises the allocation of raw transcript segments to each sub-profile at Layer 2. Two guided signals are: (i) per-construct accuracy on a holdout *calibration pool* of personas, which scores how well the current prompt's output predicts each construct, and (ii) a cumulative diff-log of prior-round revisions, which records what has already been tried.
3. **Round selection.** After rounds $0, \ldots, 4$ have been generated, we select the final reported round $r^\star$ using the *calibration-50* rule: among the five candidate rounds, choose the round with the highest calibration accuracy averaged over the 50-persona calibration pool. The full definition is given in §5.1.

## 4 BDE Structure on Twin-2K-500

We test the BDE structure on two evaluation regimes. First, on Twin-2K-500's 17 homogeneous prediction tasks, we run two paired contrasts (i.e. each persona is evaluated under each condition in the compared persona-context pair) on the same $n$=50 personas: (1) BDE versus an unstructured single-document baseline introduced in §1.1 (this comparison isolates the effect of structure) and (2) BDE versus the raw question–answer transcript (this comparison measures the joint effect of structure and compression). Second, we transfer the same BDE skeleton to the 19 pre-registered Mega-Study sub-studies Toubia et al. [2025a] — a much more heterogeneous task suite spanning various types of questions — and ask whether the within-Twin-2K-500 lift survives the shift to a heterogeneous task suite.

### 4.1 Setup and statistical framing

Let $\Omega$ denote the full Twin-2K-500 population of 2,058 personas, sampled $X_1, \ldots, X_n \sim_{\text{i.i.d.}} \Omega$. For each (configuration, metric, model) triple $(c, m, M)$—where $c$ is the persona-context configuration (BDE, unstructured summary, or raw transcript), $m$ is the accuracy metric (overall, cognitive-bias, or pricing), and $M$ is the simulator LLM (`gpt-5.4-nano`, `gpt-5.4-mini`, or `Qwen3-8B`)— write $P_{c,m,M}(X)$ for the twin's accuracy on a randomly drawn persona $X$ under that triple. We use a fixed prefix of $n$=50 paired personas reused across every prompt and contrast; the simulator is `gpt-5.4-nano` primary, with `gpt-5.4-mini` and `Qwen3-8B` replication reported in Appendix C. Pairwise contrasts are estimated by the paired sample mean

$$\widehat{\Delta} \;=\; \frac{1}{n}\sum_{i=1}^{n}\big[\,P_{c_B,m,M}(X_i) - P_{c_A,m,M}(X_i)\,\big],$$

with 95% CIs from a non-parametric paired bootstrap that resamples 50 personas with replacement over $B$=10,000 replications. This pairing exploits a strong within-persona correlation across configurations: a persona that is hard under one arm tends to be hard under the others, so the paired contrast removes persona-level baseline variation and substantially reduces variance relative to an unpaired design.

### 4.2 Findings on Twin-2K-500: BDE wins on the homogeneous task suite

Before isolating BDE structure, we anchor the unstructured summary relative to the raw transcript. Table 1 reports the unstructured-vs-raw contrast on the same $n$=50 personas. Compression alone—

going from the raw transcript to a free-form extracted summary, with no BDE structure—is roughly raw-equivalent on overall accuracy ($-0.58$pp) and pricing ($+0.40$pp), and mildly hurts cognitive bias ($-1.30$pp). The structure-vs-unstructured contrast reported next is therefore measured against a baseline already comparable to raw.

Table 1: Compression-only anchor on `gpt-5.4-nano`, paired bootstrap on $n$=50 personas. The first row is the raw transcript baseline mean accuracy; the second reports the unstructured summary mean alongside the paired $\Delta$ vs. raw with significance stars ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$; ns = 95% CI contains zero).

| Method | Overall (pp) | Cognitive bias (pp) | Pricing (pp) |
|---|---|---|---|
| Raw transcript (mean accuracy) | 69.42 | 74.20 | 63.05 |
| Unstructured ($\Delta$ vs. raw) | 68.84 ($-0.58$ ns) | 72.90 ($-1.30^{*}$) | 63.45 ($+0.40$ ns) |

A BDE structured persona produces a statistically significant lift over the unstructured summary on all three metrics (Table 2): overall $+2.49$pp (95% CI $[+1.15, +3.80]$, $p < 0.001$), cognitive bias $+2.25$pp ($[+0.98, +3.55]$, $p = 0.001$), and pricing $+2.75$pp ($[+0.68, +4.87]$, $p = 0.008$). The largest absolute gain is on pricing.

We then compare a BDE structured persona against the raw question–answer transcript that current digital-twin practice uses. On the same $n$=50 paired personas, BDE output improves over raw on all three metrics (Table 2): overall $+1.91$pp (95% CI $[+0.78, +3.11]$, $p = 0.0006$), cognitive bias $+0.95$pp ($[+0.28, +1.64]$, $p = 0.0054$), and pricing $+3.15$pp ($[+0.75, +5.70]$, $p = 0.007$). This joint contrast bundles two sources of change relative to raw: compression (raw $\rightarrow$ unstructured) and BDE structure (unstructured $\rightarrow$ BDE).

Table 2: BDE-structured contrasts on `gpt-5.4-nano`, paired bootstrap on $n$=50 personas. Paired $\Delta$ in percentage points; significance stars: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

| Contrast | Overall | Cognitive bias | Pricing |
|---|---|---|---|
| BDE-structured $-$ unstructured | $+2.49^{***}$ | $+2.25^{***}$ | $+2.75^{**}$ |
| BDE-structured $-$ raw | $+1.91^{***}$ | $+0.95^{**}$ | $+3.15^{**}$ |

### 4.3 Findings on the Mega-Study: BDE structure flattens on the heterogeneous task suite

We transfer the BDE structure that improves Twin-2K-500 performance to the 19 Mega-Study sub-studies, evaluating raw transcripts, the BDE structured persona, and an unstructured baseline with paired within-sub-study bootstrap. The BDE structured persona essentially ties the raw transcript on the 19-study aggregate (Table 3). At the sub-study level, BDE has significant ($p < 0.05$) losses relative to raw transcripts in three sub-studies. In other words, the $+1.91$pp gain on Twin-2K-500 does not transfer to the heterogeneous Mega-Study task suite. This task-conditional spread motivates the per-task auto-discovery pipeline in §5.

## 5 Auto-discovery Structure on Mega-Study

§4 showed that the BDE structured persona fails to generalize to the Mega-Study's 19 heterogeneous sub-studies. This raises the question we test in this section: for a given task, can we automatically discover a task-specific structure that outperforms the raw-transcript baseline? We use the auto-discovery method introduced in §3.3 and measure its accuracy lift over the raw transcript, BDE structured persona, and unstructured persona baselines on the locked headline metric defined in §5.1.

### 5.1 Setup

**Locked splits.** The goal of the procedure is to find a structure that generalizes across respondents on a fixed prediction task. To separate prompt refinement from final evaluation, we lock two splits for each sub-study before writing any extraction prompt. Personas are split into a development pool,

Table 3: Aggregate Mega-Study results on `gpt-5.4-nano`. Aggregation follows the Mega-Study row convention of Toubia et al. [2025a]. Deltas are percentage-point differences from the raw-transcript baseline. Significant losses are counted at $p < 0.05$ using paired-persona bootstrap within each sub-study ($B$=10,000, 50 personas per sub-study).

| Configuration | Overall (26 rows, pp) | $\Delta$ vs raw (pp) | Sig. losses vs raw |
|---|---|---|---|
| Raw questions | 71.90 | — | — |
| BDE structured | 71.90 | +0.00 | 3 / 19 |
| Unstructured | 71.88 | −0.02 | 3 / 19 |

used to generate the meta-extractor's calibration feedback, and an evaluation pool, used only for the headline metric. Predictive questions are split into calibration items, used to score candidate structures during iteration, and holdout items, used only for the reported headline accuracy.

The meta-extractor observes only construct-level aggregate accuracy on the $\text{CALIBRATION}_s \times \text{DEVELOPMENT}_s$ cell. The headline metric is computed on the doubly disjoint $\text{HOLDOUT}_s \times \text{EVAL}_s$ cell. Thus, the reported metric is separated from the meta-extractor's feedback by both a question barrier and a persona barrier. Both splits are kept frozen throughout the experiment. See Figure 3 for an illustration of the information-flow protocol.

- *Persona split*: each sub-study's 50-persona panel (§3.1) is stratified by political-party × age-bracket × education-tier and randomly partitioned into a 20-persona development pool and a 30-persona evaluation pool. The development pool is used to compute calibration feedback for prompt refinement; the evaluation pool is never observed by the meta-extractor and is used only for headline evaluation.
- *Question split*: predictive questions are stratified by construct label (§3.3) and split into calibration and holdout subsets. Calibration items are used to score candidate structures during iteration; holdout items are reserved for headline evaluation. The calibration:holdout ratio is set by the question count: 60:40 for $\geq 11$ questions, 50:50 for 4–10 questions, and *zero-shot*—all items held out, with no calibration set—for $\leq 3$ questions.

The split rule yields 13 iteration-eligible sub-studies and 6 zero-shot sub-studies whose question sets are too small to admit a calibration signal; we exclude the zero-shot sub-studies from this section's analysis (their round-0 prompts are held fixed and reported in Appendix D, Table 8) and compare on the 13 iteration-eligible sub-studies only.

**Iteration loop.** To search for a sub-study-specific structure, we run $R$=5 rounds for each iteration-eligible sub-study. Round 0 initializes the pipeline with an LLM-generated structure that is derived from the sub-study's constructs and all raw questions, while remaining blind to raw answers. This structure is used to generate an extraction prompt, which is then applied to the raw transcripts of the 20 development personas to produce their structured personas. The structured personas are passed to the simulator, which generates predicted answers for calibration questions. Per-construct calibration accuracy is then computed on the calibration questions in each construct averaged over the development persona pool.

In each of rounds 1–4, a meta-extractor LLM reads the previous rounds' per-construct calibration accuracy and the raw questions, then revises two structural choices: (i) what sub-profiles the persona should contain, including whether to add or drop sub-profiles and what information each sub-profile should include (Layer 1), and (ii) which raw questions feed each sub-profile (Layer 2). The accuracy signal is aggregated to the construct level only, with no per-persona, per-item, or holdout signal.

**Evaluation.** The selected round $r^\star$ for each sub-study is chosen by the *calibration-50* rule: we choose the round with highest accuracy on the calibration questions averaged across all 50 personas. We then use the extraction prompt from round $r^\star$ to compute the headline accuracy. In particular, the headline accuracy for sub-study $s$ is then the mean per-question score at round $r^\star$ over the locked ($\text{HOLDOUT}_s \times \text{EVAL}_s$) cells—holdout questions on the 30 evaluation personas only. Alternative selection rules and the comparison against calibration-50 are reported in Appendix E.

| | Calibration questions | Holdout questions |
|---|---|---|
| **20 development personas** | **Observed** (construct-level aggregate accuracy) | **Question barrier** never observed |
| **30 evaluation personas** | **Persona barrier** never observed | **HEADLINE** (HOLDOUT × EVAL) |

Figure 3: Information-flow protocol per construct. The orange cell (calibration questions × development personas) is the only cell the meta-extractor ever observed. Light-green cells are clean by one barrier (different questions *or* different personas than the observed cell); the dark-green cell is the headline statistic, doubly disjoint from training (different questions *and* different personas).

### 5.2 Findings and interpretation

The locked HOLDOUT × EVAL row aggregate (Table 4) directly tests whether an auto-discovered persona structure outperforms the three persona-context baselines on each sub-study. Under the row convention of Toubia et al. [2025a], the auto-discovery pipeline exceeds every baseline on the same cells, lifting mean accuracy by $+1.91$pp over raw transcripts, $+2.22$pp over BDE, and $+1.40$pp over the unstructured summary. At the sub-study level, paired within-sub-study bootstrap detects no significant losses against raw transcripts or BDE. One significant loss remains against the unstructured baseline (`infotainment`, $-3.65$pp, $p < 0.05$).

The aggregate improvement is heterogeneous rather than uniform across sub-studies. Ten of the thirteen iter-eligible sub-studies have point estimates within $\pm 3$pp of the raw-transcript baseline. With only $n{=}30$ paired evaluation personas, these near-zero contrasts are too small to distinguish reliably from sampling noise. Because holdout set sizes also vary substantially across sub-studies—from a single item to roughly 60 items—near-zero effects are statistically ambiguous rather than clear evidence of no benefit. The aggregate lift is therefore driven by a smaller set of larger wins rather than by small uniform improvements across all sub-studies.

The largest wins also arise through two different paths. Three sub-studies improve by approximately $+3$pp or more against all three baselines: `consumer_minimalism` (+12.78/+16.11/+16.11pp), `privacy` (+5.66/+7.62/+5.38pp), and `digital_certification` (+3.33/+3.33/+3.66pp). Among them, `consumer_minimalism`'s selected round is $r^{\star}{=}4$, where the iteration loop introduces a literature-anchored *Cross-Domain Consistency Profile* that is absent at round 0 (Appendix F). By contrast, `privacy` and `digital_certification` have $r^{\star}{=}0$: the LLM-generated initial structure, conditioned only on the sub-study's constructs and raw questions, already specifies a structure that no subsequent round improves under calibration-50. Thus, the aggregate lift reflects both a competitive zero-shot starting point in some sub-studies and iterative refinement in others.

## 6 Conclusion

This paper studies whether the structure of a digital-twin persona affects behavioral prediction, holding the downstream simulator fixed. On Twin-2K-500's homogeneous task family, the BDE structured persona outperforms both the unstructured persona and the raw-transcript baseline across overall, cognitive-bias, and pricing accuracy. The same BDE structure does not generalize to the heterogeneous Mega-Study task suite: on the 19-study aggregate, BDE essentially ties the raw-transcript baseline and incurs significant sub-study-level losses in three cases. Thus, the $+1.91$pp gain on Twin-2K-500 is not a universal effect of adding structure.

The auto-discovery experiment tests whether persona structure should be task-specific. On the 13 iteration-eligible Mega-Study sub-studies, the pipeline proposes one extraction prompt per sub-study, initialized from the sub-study's constructs and refined using construct-level calibration accuracy under locked development/evaluation and calibration/holdout splits. The resulting personas lift the Mega-Study overall accuracy by $+1.91$pp over raw transcripts, $+2.22$pp over BDE, and $+1.40$pp over unstructured summaries. Under paired within-sub-study bootstrap, the procedure has no significant sub-study-level losses against raw or BDE, although one significant loss remains against the unstructured baseline. Together, these results suggest that the relevant bottleneck is not only how

Table 4: Per-sub-study selected-round overall accuracy on the 13 iteration-eligible Mega-Study sub-studies (locked $\text{HOLDOUT}_s{\times}\text{EVAL}_s$ cells, $n$=30 evaluation personas/sub-study). *Sel. R* is the calibration-50-selected round; *Auto-discovery / Raw / BDE / Unstr.* are the auto-discovery, raw, BDE-structured, and unstructured headlines on the same cells. Bracketed intervals are 95% percentile CIs from paired-persona bootstrap with $B$=10,000 on the 30 evaluation personas; significance stars: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $\dagger\, 0.05 < p \leq 0.10$. Sorted by descending $\Delta$ vs raw; bottom row is the 13-study Mega-Study row-convention aggregate Toubia et al. [2025a].

| Sub-study | Sel. R | Auto. (pp) | Raw (pp) | BDE (pp) | Unstr. (pp) | $\Delta_{\text{raw}}$ (pp) [95% CI] | $\Delta_{\text{BDE}}$ (pp) [95% CI] | $\Delta_{\text{Unstr.}}$ (pp) [95% CI] |
|---|---|---|---|---|---|---|---|---|
| consumer_minimalism | R4 | 50.00 | 37.22 | 33.89 | 33.89 | $+12.78^{*}$ $[+1.11, +25.00]$ | $+16.11^{*}$ $[+3.33, +29.44]$ | $+16.11^{**}$ $[+4.44, +28.89]$ |
| privacy | R0 | 71.01 | 65.35 | 65.45 | 68.02 | $+5.66$ $[-1.98, +12.43]$ | $+5.56^{*}$ $[+1.21, +10.49]$ | $+2.99$ $[-2.01, +8.10]$ |
| digital_certification | R0 | 70.56 | 67.22 | 66.11 | 61.85 | $+3.33$ $[-5.74, +12.96]$ | $+4.44$ $[-3.05, +12.30]$ | $+8.70^{***}$ $[+3.15, +15.37]$ |
| junk_fees | R0 | 67.55 | 65.06 | 66.11 | 64.57 | $+2.49$ $[-1.05, +5.69]$ | $+1.43$ $[-1.34, +4.13]$ | $+2.98^{*}$ $[+0.45, +5.63]$ |
| context_effects | R1 | 74.17 | 71.98 | 70.94 | 76.94 | $+2.19$ $[-11.25, +16.82]$ | $+3.23$ $[-10.00, +15.10]$ | $-2.78$ $[-13.51, +5.42]$ |
| preference_redistribution | R0 | 79.21 | 77.13 | 78.57 | 78.79 | $+2.08^{\dagger}$ $[-0.42, +4.47]$ | $+0.64$ $[-2.05, +3.24]$ | $+0.41$ $[-2.09, +2.91]$ |
| accuracy_nudges | R4 | 63.37 | 61.93 | 64.73 | 66.20 | $+1.43$ $[-2.27, +5.07]$ | $-1.37$ $[-5.67, +2.93]$ | $-2.83$ $[-7.26, +0.93]$ |
| hiring_algorithms | R0 | 72.66 | 72.96 | 73.91 | 74.55 | $-0.30$ $[-2.93, +2.17]$ | $-1.25$ $[-3.47, +1.02]$ | $-1.89^{\dagger}$ $[-4.22, +0.28]$ |
| story_beliefs | R0 | 64.13 | 64.79 | 65.68 | 67.49 | $-0.66$ $[-4.06, +2.57]$ | $-1.55$ $[-4.66, +1.53]$ | $-3.36^{\dagger}$ $[-6.94, +0.31]$ |
| infotainment | R0 | 64.37 | 65.11 | 66.67 | 68.02 | $-0.75$ $[-3.25, +1.92]$ | $-2.30^{\dagger}$ $[-5.08, +0.40]$ | $-3.65^{*}$ $[-6.98, -0.48]$ |
| affective_priming | R0 | 75.77 | 76.59 | 75.28 | 75.23 | $-0.83$ $[-3.50, +2.34]$ | $+0.48$ $[-2.57, +3.81]$ | $+0.54$ $[-2.92, +4.32]$ |
| quantitative_intuition | R3 | 67.73 | 69.18 | 62.42 | 62.57 | $-1.46$ $[-10.78, +8.71]$ | $+5.31$ $[-3.41, +13.84]$ | $+5.15$ $[-3.56, +14.21]$ |
| obedient_twins | R3 | 77.64 | 80.28 | 79.45 | 80.14 | $-2.64$ $[-5.97, +0.69]$ | $-1.82$ $[-5.14, +1.62]$ | $-2.50$ $[-5.83, +0.69]$ |
| **Overall accuracy (15 rows)** | | **69.45** | 67.54 | 67.24 | 68.06 | $\mathbf{+1.91}$ | $\mathbf{+2.22}$ | $\mathbf{+1.40}$ |

much persona information is supplied, but how that information is organized: the useful structure is task-dependent.

The main practical takeaway is that when the task family is homogeneous and admits a behaviorally grounded partition, a fixed structure such as BDE is worth trying first. When tasks span heterogeneous behavioral primitives, a fixed allocation can flatten, and per-task structure discovery becomes the more appropriate design.

Several limitations remain. First, the current experiments use relatively small paired persona panels. The Twin-2K-500 contrasts use $n$=50 paired personas, while the auto-discovery experiment evaluates each sub-study on only $n$=30 paired evaluation personas. As a result, many near-zero sub-study contrasts cannot be reliably distinguished from sampling noise: their confidence intervals are wide enough to include both modest gains and modest losses. Scaling to larger evaluation panels is a direct next step. Second, we do not yet know when the calibration-50 signal is informative enough to select a useful structure. This depends not only on the number of calibration and holdout items, but also on whether those items contain predictive variation aligned with the final holdout task. With too few or weakly predictive items, round selection may mostly reflect sampling noise rather than true structure quality. Third, although we replicate the main patterns on `gpt-5.4-mini` and `Qwen3-8B`, the robustness evidence still covers only a small set of simulator families and scale points; broader cross-architecture replication remains open. Finally, our evidence is comparative rather than mechanistic: the results are consistent with the hypothesis that task-matched allocation across persona sub-profiles drives the lift, but we do not directly identify that mechanism. Future work should test alternative behavioral priors, larger persona panels, broader simulator families, and direct interventions on the allocation layer to identify which parts of the discovered structure drive the gains.

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# A Appendix: Datasets and evaluation metrics

This appendix collects dataset and evaluation details deferred from §3.1.

**Per-question scoring rule.** All accuracy metrics in this paper, on both Twin-2K-500 and the Mega-Study, score each item as

$$s_i = 1 - \frac{|y_i^{\text{twin}} - y_i^{\text{resp}}|}{r_i},$$

where $y_i^{\text{twin}}$ and $y_i^{\text{resp}}$ are the twin's and respondent's responses on question $i$, and $r_i$ is the response-scale range. Per-persona scores average items within the relevant metric or (sub-study, question-type) cell; reported numbers further average over the persona panel. The Mega-Study paper writes this as accuracy $= 1 - \text{mean}(|\text{human} - \text{twin}|)/(\text{max} - \text{min})$ Toubia et al. [2025a], which coincides with the per-item rule above when applied within question-type, where all items share a common response range.

**Twin-2K-500 cognitive-bias and pricing items.** The *cognitive-bias* metric is computed on the holdout heuristics-and-biases items in Twin-2K-500, including Linda's conjunction problem, Asian-disease framing, anchoring, sunk cost, base-rate neglect, Allais, and related items. The *pricing* metric is computed on the 40-item willingness-to-pay block, in which each item presents a branded product at a posted price and elicits a binary purchase decision. Both metrics differ only in the subset of items aggregated and therefore measure individual-level fidelity to the respondent, not aggregate-level rationality; full item wording is in Toubia et al. [2025b].

**Mega-Study persona pool per sub-study.** Unlike Twin-2K-500, where every contrast in this paper uses the same fixed set of 50 personas, the Mega-Study draws a separate persona sample of size 50 for each of the 19 sub-studies. Each sub-study's sample is drawn from the broader Twin-2K-500 persona pool, but the samples are nearly disjoint across sub-studies: the mean pairwise overlap between any two sub-studies is 1.9 personas (max 7), and the union of all 19 sub-study samples contains 695 unique personas out of $50 \times 19 = 950$ possible slots.

**Mega-Study outcome set per sub-study.** Each sub-study's holdout outcome set has its own item count and item-type mix. Question counts in our setup span 1 (`targeting_fairness` on the single targeting-judgment item) to $\sim 60+$ (`junk_fees` on the seven-domain fairness $\times$ familiarity grid), with mean $\sim 16$. Item types include six- and seven-point Likert multiple choice, agreement matrices, sliders, and multi-select platform-use items. The same range-normalized accuracy formula is used for every Mega-Study configuration we report, so all baselines are on the same scale.

**Mega-Study aggregation conventions.** The Mega-Study paper aggregates across sub-studies at the *(study, DV-var) row* level: each sub-study contributes one row per dependent variable to a long-format table, and the aggregate "mean accuracy" is the unweighted mean over rows Toubia et al. [2025a].[1] We adopt this row convention as the default aggregation rule for every Mega-Study table in this paper (Tables 3, 4, 6, and 7) so that our headline aggregates are directly comparable to Toubia et al. [2025a]'s headline tables. On `gpt-5.4-nano` for the Section 4 heterogeneity contrast, raw row-mean accuracy is 0.7190 over the 26 (sub-study, DV-var) rows spanning the 19 sub-studies (the strict `mc_exact` metric is excluded, matching what Toubia et al. [2025a]'s aggregator ingests); the structured-minus-raw gap is $+0.00$pp and the unstructured-minus-raw gap is $-0.02$pp under the row convention. For the Section 5 iteration tables, the row count drops to 15 rows over the 13 iteration-eligible sub-studies because the headline cell is restricted to the locked HOLDOUT $\times$ EVALUATION grid: 11 studies contribute one MC row, while `hiring_algorithms` and `privacy` each contribute one MC and one Matrix row. Reproducibility scripts: `evaluation/row_convention_aggregate.py` (Section 4), `evaluation/holdout_row_from_accuracy_json.py` (Section 5, item-level row aggregation against the per-study accuracy JSON files in `Digital-Twin-Simulation/text_simulation/accuracy/`).

[1] See `mega_study_evaluation/create_summary_table.py` in the Mega-Study release: the aggregator is `df.groupby("persona specification")[metrics].mean()`, which weights each row equally and therefore gives more weight to sub-studies with more DV-vars.

**Use across experiments.** §4 contrasts raw transcripts against hand-crafted BDE on all 19 sub-studies. §5 contrasts the selected round of the iteration pipeline against raw and structured baselines on the 13 iteration-eligible sub-studies; the 6 zero-shot sub-studies, with $\leq 3$ predictive items, are reported separately in Appendix D. All bootstraps are paired within sub-study with $B{=}10{,}000$ resamples. The primary simulator is `gpt-5.4-nano`; all robustness analyses are repeated on `gpt-5.4-mini` and `Qwen3-8B` (thinking) in §C.

# B Appendix: BDE representation details

This appendix illustrates the structured representation introduced in §3.2. At simulation time the LLM receives the three section files (`background.txt`, `decision_procedure.txt`, `evaluation_profile.txt`) concatenated under section headers, totalling roughly 250 lines of prose per persona, followed by the test question. The blocks below are short *excerpts* from one respondent (`pid_4`) reproduced verbatim from the extracted persona; values, prose, and demographic categories are unchanged. They illustrate the format the simulator sees, not the full input.

**Background (`bg`) — demographic block (full).**

> This individual is a White male aged between 50 and 64, residing in the Southern United States. He is a U.S. citizen, married, and currently employed full-time. His highest education level is college graduate or some postgraduate education. His household consists of two people, likely himself and his spouse. His annual family income falls between \$75,000 and \$100,000, indicating a comfortable middle-class economic status. He identifies his religion as “Other” and reports never attending religious services outside of weddings and funerals. Politically, he identifies as a very liberal Democrat, indicating strong progressive political views. These demographic details suggest he likely has stable employment and moderate time availability, with a household structure that may allow for focused personal and professional development.

**Decision procedure (`dp`) — effort-regulation excerpt.**

> **Effort regulation and cognitive style**
> This individual exhibits a strong preference for engaging in effortful cognitive activity, as evidenced by high Need for Cognition (NFC) scores: they “like to have the responsibility of handling a situation that requires a lot of thinking” (5 — strongly agree), “really enjoy a task that involves coming up with new solutions” (5), and “prefer intellectual, difficult, and important tasks” (5). They strongly disagree with statements indicating avoidance of thinking or mental effort, such as “thinking is not my idea of fun” (1) and “I only think as hard as I have to” (1). This aligns with their Big Five profile showing very high Conscientiousness (e.g., “does a thorough job” 5, “is a reliable worker” 5, “perseveres until the task is finished” 5) and high Openness to Experience.
>
> Cognitive test performance is mixed but generally competent. There is a notable error in the race question (“If you pass the person in second place, what place are you in?” answered as 1 instead of 2), and the bat-and-ball problem was answered incorrectly (ball cost = \$0.05 expected, answered \$5), a classic cognitive-reflection failure. Overall the participant shows a strong analytical style with occasional lapses in cognitive reflection.

**Evaluation profile (`ep`) — risk-preferences excerpt.**

> **Risk preferences**
> Lottery and risk tasks show moderate risk aversion, especially in gains, with a preference for certainty over risky lotteries unless the sure amount is close to the expected value. The person values reducing risk and uncertainty, consistent with high conscientiousness and a preference for order and predictability. Loss aversion is evident, with a strong preference to avoid losses and reject lotteries involving potential losses even when expected value is favorable. The person’s cognitive style (high conscientiousness, dislike of uncertainty) supports risk-averse behavior, likely leading to cautious product choices and reluctance to try unfamiliar brands or novel products unless they are low risk or well vetted. Confidence in this inference is high.

# C Appendix: Robustness across LLMs

The hand-crafted track on Twin-2K-500 (§4–§4.2) and the auto-discovery pipeline on the Mega-Study (§5) are both run on `gpt-5.4-nano` as primary. We replicate both on `gpt-5.4-mini` and on `Qwen3-8B` (thinking). §C.1 covers the hand-crafted BDE design; §C.2 covers the calibration-50 selected extraction prompts produced by the auto-discovery pipeline.

## C.1 BDE-structured design on Twin-2K-500

For each replication model we report the BDE-structured design under two paired contrasts: against the unstructured-summary baseline (the structural contribution isolated in §4) and against the raw-transcript baseline (the end-to-end gain over the standard digital-twin recipe). All other factors are held identical to the primary nano experiments: the same $n$=50 paired personas, the same hand-crafted BDE allocation, and the same paired bootstrap with $B$=10,000 resamples.

Table 5: Cross-model robustness of the BDE-structured design on Twin-2K-500. For each replication model, paired-bootstrap $\Delta$ (in percentage points) of BDE-structured against two baselines: the unstructured-summary baseline and the raw-transcript baseline. $n$=50 paired personas, $B$=10,000 resamples. Significance stars: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$; ns = 95% CI contains zero.

| Model | Contrast | Overall | Cognitive bias | Pricing |
|---|---|---|---|---|
| `gpt-5.4-mini` | vs. unstructured | $+1.27^{*}$ | $+0.76$ ns | $+1.95$ ns |
| `gpt-5.4-mini` | vs. raw transcript | $+2.43^{***}$ | $+1.76^{***}$ | $+3.15$ ns |
| `Qwen3-8B` (thinking) | vs. unstructured | $+2.61^{***}$ | $+0.03$ ns | $+5.90^{***}$ |
| `Qwen3-8B` (thinking) | vs. raw transcript | $+1.04$ ns | $+2.40^{***}$ | $-0.75$ ns |

**Findings.** On both replication models, BDE-structured output improves *overall* accuracy over the unstructured-summary baseline ($+1.27^{*}$pp on `gpt-5.4-mini`; $+2.61^{***}$pp on `Qwen3-8B`), reproducing the primary nano contrast in direction and significance. The end-to-end contrast against the raw-transcript baseline is sharper on `gpt-5.4-mini` ($+2.43^{***}$pp overall, with $+1.76^{***}$pp on cognitive bias and a borderline $+3.15$pp on pricing, $p = 0.053$) and weaker on `Qwen3-8B` (overall $+1.04$pp ns; the gain concentrates in cognitive bias, $+2.40^{***}$pp). The per-metric breakdown remains model-specific: on `gpt-5.4-mini` every metric is directionally positive on both contrasts, while on `Qwen3-8B` pricing is the largest effect against unstructured ($+5.90^{***}$pp) but does not transfer against raw. The overall pattern—the BDE-vs-unstructured contrast holds in direction and significance across both replication models, while per-metric effects vary by model and baseline—supports the structural contribution as a robust design choice rather than a model-specific artefact.

## C.2 Calibration-50 selected extraction prompts on the Mega-Study

The calibration-50 selection rule chooses the auto-discovery pipeline's selected round purely on the `gpt-5.4-nano` simulator's calibration accuracy (§5.1); the resulting extraction-prompt files are artefacts of that pipeline that should, in principle, be re-usable as persona context for any sufficiently capable simulator. We test this by swapping the simulator to Qwen3-8B (thinking mode, served via the Dashscope OpenAI-compatible endpoint) and to `gpt-5.4-mini` (thinking, reasoning=high), re-running the iteration-eligible sub-studies on the same locked $(\text{HOLDOUT}_s \times \text{EVALUATION}_s)$ cells. Four persona-context conditions are compared per simulator (Tables 6 and 7): *Calibration-50* re-uses the calibration-50-selected extraction prompts from the auto-discovery pipeline unchanged; *Raw* substitutes the raw past-survey transcript; *Str.* uses the hand-crafted BDE summary; and *Unstr.* uses the unstructured-summary baseline. The structured-JSON answer schema is identical across all four, so each contrast isolates persona-context format. Under the Mega-Study row convention of Toubia et al. [2025a] (one row per (sub-study, DV-var) cell, 15 rows), Calibration-50 leads on both simulators: on Qwen3-8B the row aggregates are 65.98 (Calibration-50), 62.73 (Raw), 64.20 (Str.), 63.25 (Unstr.) with $\Delta_{\text{calibration-raw}} = +3.26$pp, $\Delta_{\text{calibration-str.}} = +1.78$pp, $\Delta_{\text{calibration-unstr.}} = +2.73$pp; on `gpt-5.4-mini` the corresponding row aggregates are 69.71, 68.70, 69.67, 69.66 with $\Delta_{\text{calibration-raw}} = +1.01$pp, $\Delta_{\text{calibration-str.}} = +0.04$pp, $\Delta_{\text{calibration-unstr.}} = +0.05$pp (point estimates; same-sign mirrors of the $+1.91$ and $+2.22$ effects on gpt-5.4-nano in Table 4). Tables 6 and 7 include 95% CIs and significance markers from paired-persona bootstrap with $B$=10,000 on the same 30 evaluation personas. The auto-discovery pipeline's calibration-50 effect therefore transfers directionally to both a smaller open-weights simulator and to a different OpenAI scale-point, with attenuated magnitude on the larger gpt-5.4-mini; the rerun scripts are released under `iteration/run_qwen_eval30.py`, `iteration/score_unstructured_eval30.py`, and the mini-driver `Digital-Twin-Simulation/text_simulation/run_mini_megastudy_qwen_eval30.sh`.

Table 6: Iteration-pipeline robustness on Qwen3-8B (thinking). Qwen3-8B is swapped in for gpt-5.4-nano as the simulator on the 13 iteration-eligible Mega-Study sub-studies (sort matches Table 4). *Calibration-50* re-uses the calibration-50-selected extraction prompts unchanged; *Raw* substitutes the raw past-survey transcript; *Str.* uses the hand-crafted BDE summary; *Unstr.* uses the unstructured-summary baseline. All four columns are scored on the same locked (HOLDOUT $\times$ EVAL) cells with $n$=30 evaluation personas. Bracketed intervals are 95% percentile CIs from paired-persona bootstrap with $B$=10,000; significance stars: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $\dagger\, 0.05 < p \leq 0.10$. The bottom row aggregates under the Mega-Study row convention of Toubia et al. [2025a] (one row per (sub-study, DV-var) cell, 15 rows over the 13 iteration-eligible sub-studies).

| Sub-study | Sel. | Calibration-50 (pp) | Raw (pp) | Str. (pp) | Unstr. (pp) | $\Delta_{\text{calibration-raw}}$ (pp) [95% CI] | $\Delta_{\text{calibration-str.}}$ (pp) [95% CI] | $\Delta_{\text{calibration-unstr.}}$ (pp) [95% CI] |
|---|---|---|---|---|---|---|---|---|
| consumer_minimalism | R4 | 52.22 | 33.33 | 35.56 | 26.11 | $+18.89^{*}$ $[+1.67, +36.67]$ | $+16.67^{\dagger}$ $[-0.56, +33.89]$ | $+26.11^{***}$ $[+12.22, +41.11]$ |
| privacy | R0 | 62.78 | 54.20 | 60.80 | 58.16 | $+8.58^{*}$ $[+1.39, +19.60]$ | $+1.98$ $[-5.94, +9.22]$ | $+4.62$ $[-3.99, +13.61]$ |
| digital_certification | R0 | 62.04 | 57.78 | 55.00 | 52.59 | $+4.26$ $[-3.70, +12.41]$ | $+7.04^{*}$ $[+0.56, +14.71]$ | $+9.44^{**}$ $[+2.58, +17.00]$ |
| junk_fees | R0 | 63.78 | 61.34 | 61.22 | 61.21 | $+2.44$ $[-1.13, +6.12]$ | $+2.56^{*}$ $[+0.04, +5.34]$ | $+2.57$ $[-1.50, +6.50]$ |
| context_effects | R1 | 68.54 | 66.35 | 69.48 | 71.88 | $+2.19$ $[-7.19, +13.75]$ | $-0.94$ $[-14.48, +12.62]$ | $-3.33$ $[-11.94, +6.43]$ |
| preference_redistribution | R0 | 80.68 | 78.61 | 80.42 | 79.96 | $+2.08^{*}$ $[+0.07, +3.99]$ | $+0.26$ $[-1.77, +2.43]$ | $+0.72$ $[-1.46, +2.84]$ |
| accuracy_nudges | R4 | 60.20 | 58.90 | 60.27 | 63.50 | $+1.30$ $[-5.33, +8.73]$ | $-0.07$ $[-7.75, +7.96]$ | $-3.30$ $[-11.50, +3.13]$ |
| hiring_algorithms | R0 | 69.01 | 66.17 | 67.45 | 67.20 | $+2.84$ $[-2.26, +8.15]$ | $+1.55$ $[-1.92, +5.08]$ | $+1.81$ $[-1.13, +4.98]$ |
| story_beliefs | R0 | 68.26 | 70.00 | 69.50 | 69.51 | $-1.74$ $[-5.49, +2.15]$ | $-1.24$ $[-4.74, +2.29]$ | $-1.25$ $[-4.10, +1.46]$ |
| infotainment | R0 | 66.27 | 66.35 | 67.70 | 66.19 | $-0.08$ $[-4.37, +4.29]$ | $-1.43$ $[-7.38, +4.52]$ | $+0.08$ $[-5.87, +5.87]$ |
| affective_priming | R0 | 62.55 | 66.06 | 66.50 | 67.87 | $-3.51$ $[-10.08, +2.84]$ | $-3.95$ $[-9.67, +1.51]$ | $-5.32$ $[-11.86, +1.60]$ |
| quantitative_intuition | R3 | 64.25 | 67.13 | 62.40 | 61.24 | $-2.88$ $[-9.62, +4.51]$ | $+1.85$ $[-6.92, +10.30]$ | $+3.01$ $[-8.62, +13.06]$ |
| obedient_twins | R3 | 77.36 | 74.28 | 78.47 | 78.02 | $+3.08^{\dagger}$ $[-0.39, +6.67]$ | $-1.11$ $[-4.17, +1.81]$ | $-0.66$ $[-4.72, +3.38]$ |
| **Row mean (15 rows)** | | **65.98** | 62.73 | 64.20 | 63.25 | **+3.26** | **+1.78** | **+2.73** |

## D Appendix: Iteration pipeline implementation and supplementary results

This appendix specifies the iteration pipeline used in §5 and reports supplementary results not in the main text.

**Output-barrier validator.** Beyond the question/persona disjointness shown in Figure 3 and the construct-level calibration aggregation already noted in §5.1, each meta-extractor proposal passes through a 9-rule fail-closed validator before it is saved as the next-round extraction prompt. A proposal is rejected and re-asked (up to 2 retries; falls back to the prior round on the third failure) if any of the following nine checks fail:

1. The `{transcript}` placeholder is missing.
2. The round-0 output marker is altered.
3. Any `Q[0-9]+` regex matches inside the output section (Q-number leakage).

Table 7: Iteration-pipeline robustness on `gpt-5.4-mini` (thinking, reasoning=high). The simulator is swapped for `gpt-5.4-nano` on the 13 iteration-eligible Mega-Study sub-studies (sort matches Table 4). *Calibration-50* re-uses the calibration-50-selected extraction prompts unchanged; *Raw* substitutes the raw past-survey transcript; *Str.* uses the hand-crafted BDE summary; *Unstr.* uses the unstructured-summary baseline. All four columns are scored on the same locked (HOLDOUT $\times$ EVAL) cells with $n$=30 evaluation personas. Bracketed intervals are 95% percentile CIs from paired-persona bootstrap with $B$=10,000; significance stars: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $\dagger\, 0.05 < p \leq 0.10$. The bottom row aggregates under the Mega-Study row convention of Toubia et al. [2025a] (one row per (sub-study, DV-var) cell, 15 rows over the 13 iteration-eligible sub-studies).

| Sub-study | Sel. | Calibration-50 (pp) | Raw (pp) | Str. (pp) | Unstr. (pp) | $\Delta_{\text{calibration-raw}}$ (pp) [95% CI] | $\Delta_{\text{calibration-str.}}$ (pp) [95% CI] | $\Delta_{\text{calibration-unstr.}}$ (pp) [95% CI] |
|---|---|---|---|---|---|---|---|---|
| `consumer_minimalism` | R4 | 43.89 | 29.44 | 32.78 | 30.00 | $+14.44^{\dagger}$ $[-0.56, +28.89]$ | $+11.11$ $[-3.33, +25.00]$ | $+13.89^{\dagger}$ $[-1.11, +28.33]$ |
| `privacy` | R0 | 67.12 | 74.79 | 72.71 | 70.49 | $-7.67^{***}$ $[-11.66, -3.61]$ | $-5.59^{**}$ $[-9.62, -1.34]$ | $-3.37^{\dagger}$ $[-7.44, +0.41]$ |
| `digital_certification` | R0 | 70.56 | 74.81 | 67.04 | 71.30 | $-4.26$ $[-13.62, +5.19]$ | $+3.52$ $[-5.08, +12.41]$ | $-0.74$ $[-6.19, +4.82]$ |
| `junk_fees` | R0 | 72.87 | 71.66 | 72.67 | 71.52 | $+1.21$ $[-0.26, +2.82]$ | $+0.19$ $[-1.71, +1.97]$ | $+1.35$ $[-0.62, +3.59]$ |
| `context_effects` | R1 | 66.88 | 62.81 | 63.75 | 71.77 | $+4.06$ $[-10.42, +17.14]$ | $+3.12$ $[-14.01, +18.61]$ | $-4.90$ $[-17.20, +6.67]$ |
| `preference_redistribution` | R0 | 81.69 | 81.28 | 81.41 | 82.37 | $+0.42$ $[-1.54, +2.44]$ | $+0.28$ $[-1.16, +1.87]$ | $-0.67$ $[-2.46, +1.12]$ |
| `accuracy_nudges` | R4 | 68.07 | 63.87 | 71.93 | 66.80 | $+4.20$ $[-0.87, +8.40]$ | $-3.87^{*}$ $[-6.60, -0.00]$ | $+1.27$ $[-3.33, +6.40]$ |
| `hiring_algorithms` | R0 | 72.89 | 73.71 | 74.05 | 74.03 | $-0.82$ $[-2.84, +1.14]$ | $-1.15$ $[-3.01, +0.73]$ | $-1.13$ $[-3.17, +0.79]$ |
| `story_beliefs` | R0 | 69.69 | 68.12 | 68.23 | 67.78 | $+1.56$ $[-0.97, +4.17]$ | $+1.46$ $[-1.67, +4.76]$ | $+1.91$ $[-0.83, +4.67]$ |
| `infotainment` | R0 | 71.03 | 68.25 | 71.03 | 72.46 | $+2.78$ $[-0.95, +6.35]$ | $-0.00$ $[-3.49, +3.33]$ | $-1.43$ $[-4.44, +1.51]$ |
| `affective_priming` | R0 | 74.61 | 72.98 | 77.00 | 77.48 | $+1.63$ $[-4.35, +7.63]$ | $-2.38$ $[-6.23, +1.14]$ | $-2.87$ $[-6.75, +0.63]$ |
| `quantitative_intuition` | R3 | 66.03 | 61.34 | 64.57 | 64.61 | $+4.69$ $[-2.02, +12.73]$ | $+1.46$ $[-7.22, +10.74]$ | $+1.42$ $[-6.42, +9.52]$ |
| `obedient_twins` | R3 | 80.28 | 78.89 | 81.11 | 79.72 | $+1.39$ $[-1.81, +4.58]$ | $-0.83$ $[-4.03, +2.22]$ | $+0.56$ $[-2.22, +3.19]$ |
| **Row mean (15 rows)** | | **69.71** | 68.70 | 69.67 | 69.66 | **+1.01** | **+0.04** | **+0.05** |

4. Any “Predicted Answer Pattern” or per-item prediction phrase appears.
5. Either of the required output-section headers (`## Behavioral Disposition Summary`, `## Self-Report Reliability Note`) is removed.
6. The sub-profile count falls outside $[2, 6]$.
7. Fewer than 3 distinct schema modules from the 500-question Twin-2K-500 transcript are referenced.
8. The sub-study’s primary-anchor keyword is dropped.
9. Any documented failure-mode signature from a prior round is removed (additive-only rule).

**Cross-round meta-extractor context.** §5.1 specifies the per-round meta-extractor input. From round 2 onward the meta-extractor additionally receives the verbatim `DIFF_LOG` blocks from prior rounds, an “already-addressed constructs” table, and a $\uparrow$ / $\downarrow$ / $\sim$ trend column on the per-construct calibration trajectory. The template forbids re-fixing a construct already targeted by an earlier round unless its trajectory shows an unambiguous regression, and forbids reverting a fix unless its trajectory shows the fix harmed accuracy. These rules block the duplicate-fix and round-to-round-oscillation failure modes observed in pilots.

**Simulator non-determinism.** The simulator (`gpt-5.4-nano` at `reasoning_effort=high`) is invoked through the OpenAI Responses API, which does not accept `temperature` or `seed` for reasoning-effort models in this family (verified empirically: setting `temperature=0` returns HTTP 400 with “Only the default (1) value is supported”). The simulator therefore runs at the model’s fixed default sampling, which contributes a non-trivial round-to-round variance even when the extraction prompt is byte-identical. We do not control for this beyond the locked splits and the calibration-50 selection rule's $\pm 0.002$ tie-break to the earliest round.

**Zero-shot sub-studies.** Table 8 reports round-0 headline accuracies for the 6 zero-shot sub-studies excluded from Table 4 (§5.1). Because their round-0 prompt is held fixed across rounds, any round-to-round movement is by construction simulator sampling noise on a byte-identical prompt; selecting a "best round" on this subset would inflate the aggregate.

Table 8: Zero-shot sub-studies (no calibration items; round-0 extraction prompt held fixed across rounds 0–4). Round-0 headline accuracy on the locked ($\text{HOLDOUT}_s \times \text{EVALUATION}_s$) cells, vs. raw and structured baselines on the same cells. These sub-studies are excluded from the main-text aggregate (Table 4) because the iteration procedure does not act on them. †`idea_generation` is an open-ended creativity task set (divergent thinking, alternative-uses, generate-an-idea) whose items are free-text and not scored by the JSON-direct parser used elsewhere in this table; the 100.0 values reflect a single screening / consent MC item that all conditions answer correctly, and should not be read as a substantive creativity score.

| Sub-study | R0 headline (pp) | raw (pp) | structured (pp) |
|---|---|---|---|
| `default_eric` | 59.6 | 62.1 | 65.4 |
| `idea_evaluation` | 58.0 | 83.9 | 85.3 |
| `idea_generation`† | 100.0 | 100.0 | 100.0 |
| `promiscuous_donors` | 75.6 | 73.4 | 74.8 |
| `recommendation_algorithms` | 74.4 | 73.3 | 72.2 |
| `targeting_fairness` | 70.4 | 75.0 | 76.7 |

**Per-round accuracy trajectories.** Figure 4 plots, for each of the 13 iteration-eligible sub-studies, the per-round headline (hold $\times$ evaluation), calibration (20-persona), and holdout (30-persona) trajectories across rounds 0–4. The 6 zero-shot sub-studies are omitted because their prompts are byte-identical across all rounds (no iteration applied), so the per-round trajectory carries no signal about the iteration procedure.

**Released artefacts.** The pipeline ships:

1. The 13 final per-sub-study extraction prompts under `iteration/final/final_extraction_<study>.md`.
2. The round-by-round prompts under `iteration/round{0..4}/round{N}_extraction_<study>.md`.
3. The meta-extractor prompt template under `iteration/iteration_subagent_prompt_template.md`.
4. The 9-rule validator under `iteration/iteration_validator.py`.
5. The paired-bootstrap CSVs under `iteration/logs/{final_vs_raw, paired_bootstrap_converged}.csv`.

A practitioner deploying the pipeline pays one extractor pass per (persona, sub-study) cell at evaluation time; the iterative refinement loop is upstream tooling and not part of the deployment cost.

# E Appendix: Selection-rule comparison

The calibration-50 rule introduced in §5.1 is one choice among several plausible round-selection policies. This appendix benchmarks calibration-50 against two alternatives on the same locked ($\text{HOLDOUT}_s \times \text{EVAL}_s$) cells used by Table 4.

- **calibration-50** (paper default): the round with highest mean accuracy on calibration questions averaged over all 50 personas, ties within $\pm 0.002$ broken to the earliest round.
- **calibration-20**: the round with highest mean accuracy on calibration questions averaged over 20 personas from the development pool, ties within $\pm 0.002$ broken to the earliest round.

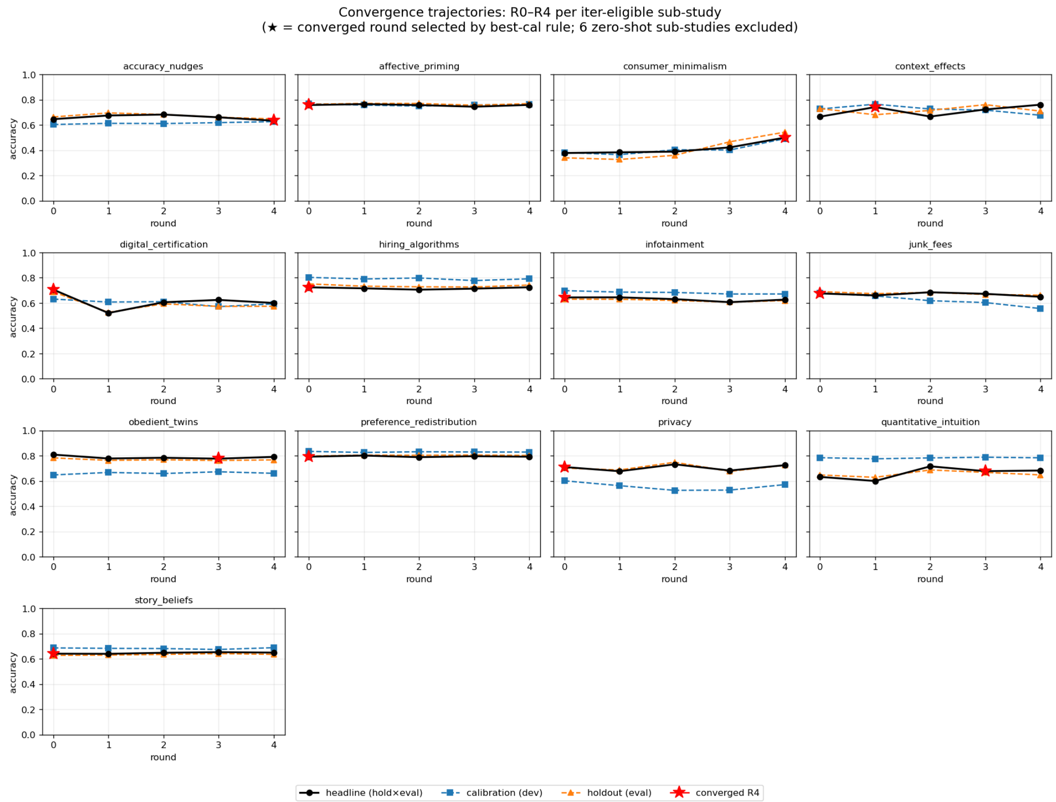


Figure 4: Per-round accuracy trajectories for the iteration pipeline, rounds 0–4, on the 13 iteration-eligible Mega-Study sub-studies. Black solid: headline accuracy on (HOLDOUT $\times$ EVAL). Blue dashed: calibration accuracy on (calibration items $\times$ all 50 personas), the calibration-50 selection statistic. Orange dashed: holdout accuracy on (holdout items $\times$ all 50 personas), untouched by selection. Red star: selected round $r^\star$ (calibration-50). The 6 zero-shot sub-studies (no calibration items, prompts byte-identical across rounds) are omitted from this overview; their round-0 headlines are tabulated in Table 8.

- **oracle** (general-best): the round with highest mean headline accuracy on HOLDOUT $\times$ EVAL itself. This rule peeks at the headline cell and is therefore not deployable; it is reported only as a power upper bound.

The per-sub-study selected round and headline accuracy at that round under each rule are reported in Tables 9 and 10 (both styled to match Table 4). Three patterns are notable.

*Calibration-50 captures a sizeable fraction of the oracle headroom.* Under the row convention of Toubia et al. [2025a], the oracle's row-mean headline of 70.91pp is the upper bound on what any round-selection rule could deliver from the locked five-round trajectory. Calibration-50 reaches 69.45pp, recovering $\approx 57\%$ of the oracle's lift over raw (+1.91 vs +3.37pp). The remaining gap concentrates on sub-studies whose calibration signal plateaus while the headline cell still moves (`privacy`, `quantitative_intuition`, `story_beliefs`, `obedient_twins`), consistent with the construct-level calibration aggregate being a lossy proxy for per-question, per-persona holdout structure.

*Cal-20 underperforms calibration-50 and falls below the unstructured baseline.* The dev-only signal is noisier: restricted to 20 personas, the calibration accuracy ranking of rounds is overwhelmed by sampling noise on small calibration constructs, so the rule selects late rounds whose holdout performance regressed. Two failure modes are diagnostic: `digital_certification` (calibration-50 picks $r^\star$=0 at 70.56pp; calibration-20 picks $r^\star$=1 at 52.04pp, a $-18.5$pp drop), and `context_effects` (calibration-50 $r^\star$=1 at 74.17pp; calibration-20 $r^\star$=2 at 66.67pp, a $-7.5$pp drop). Aggregated, calibration-20's row-mean headline is 68.12pp ($\Delta_{\text{raw}} = +0.58$pp,

$\Delta_{\text{BDE}} = +0.88$pp, $\Delta_{\text{Unstr.}} = +0.06$pp), confirming that calibration-50's earliest-round tie-break and inclusion of all 50 personas function as guards against development-calibration overfit.

*Selected-round distributions diverge.* Calibration-50 keeps round-0 on $8/13$ sub-studies, calibration-20 on $5/13$, and the oracle on $4/13$; the remaining mass for calibration-20 and the oracle is shifted toward middle rounds ($r^\star = 1$ or 2) rather than late rounds. A reading consistent with the per-sub-study trace is that the calibration signal becomes informative (and the meta- extractor's revisions stop being random) only on a minority of sub-studies; on the rest, picking the earliest valid round is not a loss-bearing default.

Table 9: Per-sub-study selected round and headline accuracy on the 13 iteration-eligible Mega-Study sub-studies under the *calibration-20* selection rule (locked $\textsc{Holdout}_s \times \textsc{Eval}_s$, $n$=30 evaluation personas / sub-study). *Sel. R* is the calibration-20-selected round; *Round / Raw / BDE / Unstr.* are the round-$r^\star$, raw, BDE, and unstructured-summary headline accuracies on the same cells. Bracketed intervals are 95% percentile CIs from paired-persona bootstrap with $B$=10,000; significance stars: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $\dagger\, 0.05 < p \leq 0.10$. Sorted to match Table 4; bottom row is the Mega-Study row-convention aggregate Toubia et al. [2025a].

| Sub-study | Sel. R | Round (pp) | Raw (pp) | BDE (pp) | Unstr. (pp) | $\Delta_{\text{raw}}$ (pp) [95% CI] | $\Delta_{\text{BDE}}$ (pp) [95% CI] | $\Delta_{\text{Unstr.}}$ (pp) [95% CI] |
|---|---|---|---|---|---|---|---|---|
| `consumer_minimalism` | R4 | 50.00 | 37.22 | 33.89 | 33.89 | $+12.78^{*}$ $[+1.11, +25.00]$ | $+16.11^{*}$ $[+3.33, +29.44]$ | $+16.11^{**}$ $[+4.44, +28.89]$ |
| `privacy` | R0 | 71.01 | 65.35 | 65.45 | 68.02 | $+5.66$ $[-1.98, +12.43]$ | $+5.56^{*}$ $[+1.21, +10.49]$ | $+2.99$ $[-2.01, +8.10]$ |
| `digital_certification` | R1 | 52.04 | 67.22 | 66.11 | 61.85 | $-15.19^{***}$ $[-23.08, -7.13]$ | $-14.07^{**}$ $[-23.33, -4.81]$ | $-9.81^{*}$ $[-18.94, -1.14]$ |
| `junk_fees` | R0 | 67.55 | 65.06 | 66.11 | 64.57 | $+2.49$ $[-1.05, +5.69]$ | $+1.43$ $[-1.34, +4.13]$ | $+2.98^{*}$ $[+0.45, +5.63]$ |
| `context_effects` | R2 | 66.67 | 71.98 | 70.94 | 76.94 | $-5.31$ $[-21.13, +12.68]$ | $-4.27$ $[-18.15, +9.58]$ | $-10.28^{\dagger}$ $[-21.38, +1.11]$ |
| `preference_redistribution` | R3 | 79.58 | 77.13 | 78.57 | 78.79 | $+2.45^{*}$ $[+0.26, +4.60]$ | $+1.01$ $[-1.66, +3.64]$ | $+0.78$ $[-1.64, +3.11]$ |
| `accuracy_nudges` | R1 | 67.50 | 61.93 | 64.73 | 66.20 | $+5.57^{\dagger}$ $[-0.82, +8.90]$ | $+2.77$ $[-2.96, +6.80]$ | $+1.30$ $[-6.19, +5.53]$ |
| `hiring_algorithms` | R0 | 72.66 | 72.96 | 73.91 | 74.55 | $-0.30$ $[-2.93, +2.17]$ | $-1.25$ $[-3.47, +1.02]$ | $-1.89^{\dagger}$ $[-4.22, +0.28]$ |
| `story_beliefs` | R0 | 64.13 | 64.79 | 65.68 | 67.49 | $-0.66$ $[-4.06, +2.57]$ | $-1.55$ $[-4.66, +1.53]$ | $-3.36^{\dagger}$ $[-6.94, +0.31]$ |
| `infotainment` | R0 | 64.37 | 65.11 | 66.67 | 68.02 | $-0.75$ $[-3.25, +1.92]$ | $-2.30^{\dagger}$ $[-5.08, +0.40]$ | $-3.65^{*}$ $[-6.98, -0.48]$ |
| `affective_priming` | R1 | 76.53 | 76.59 | 75.28 | 75.23 | $-0.06$ $[-2.46, +2.63]$ | $+1.25$ $[-1.34, +4.26]$ | $+1.30$ $[-1.46, +4.42]$ |
| `quantitative_intuition` | R3 | 67.73 | 69.18 | 62.42 | 62.58 | $-1.46$ $[-10.78, +8.71]$ | $+5.31$ $[-3.41, +13.84]$ | $+5.15$ $[-3.56, +14.21]$ |
| `obedient_twins` | R2 | 78.33 | 80.28 | 79.45 | 80.14 | $-1.94$ $[-4.58, +0.83]$ | $-1.12$ $[-4.03, +2.11]$ | $-1.81$ $[-4.86, +1.39]$ |
| **Row mean (15 rows)** | | **68.12** | 67.54 | 67.24 | 68.06 | **+0.58** | **+0.88** | **+0.06** |

**Selected-round distribution.**

| Rule | R0 | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|
| calibration-50 | 8 | 1 | 0 | 2 | 2 |
| calibration-20 | 5 | 3 | 2 | 2 | 1 |
| oracle | 4 | 2 | 4 | 1 | 2 |

The selection-rule trace is released under `iteration/logs/selection_rule_comparison.csv`.

## F Appendix: Examples of selected-round structures

This appendix surfaces the top-level sub-profile names that the iteration pipeline produced at each iteration-eligible sub-study's selected round $r^\star$ (§5.1), plus the round-0 LLM-generated structures used unchanged on the 6 zero-shot sub-studies. Table 11 lists the names verbatim;

Table 10: Per-sub-study selected round and headline accuracy on the 13 iteration-eligible Mega-Study sub-studies under the *oracle* (general-best) selection rule, which picks the round with highest mean headline accuracy on the locked $\text{HOLDOUT}_s \times \text{EVAL}_s$ cells. The rule peeks at the reported metric and is therefore not deployable; it is reported only as a power upper bound on what any round-selection rule could deliver from the locked five-round trajectory. Columns, bootstrap convention, and sort order match Table 9.

| Sub-study | Sel. R | Round (pp) | Raw (pp) | BDE (pp) | Unstr. (pp) | $\Delta_{\text{raw}}$ (pp) [95% CI] | $\Delta_{\text{BDE}}$ (pp) [95% CI] | $\Delta_{\text{Unstr.}}$ (pp) [95% CI] |
|---|---|---|---|---|---|---|---|---|
| `consumer_minimalism` | R4 | 50.00 | 37.22 | 33.89 | 33.89 | $+12.78^{*}$ $[+1.11, +25.00]$ | $+16.11^{*}$ $[+3.33, +29.44]$ | $+16.11^{**}$ $[+4.44, +28.89]$ |
| `privacy` | R2 | 73.15 | 65.35 | 65.45 | 68.02 | $+7.80^{**}$ $[+2.36, +12.92]$ | $+7.70^{***}$ $[+3.82, +11.99]$ | $+5.13^{*}$ $[+0.31, +10.18]$ |
| `digital_certification` | R0 | 70.56 | 67.22 | 66.11 | 61.85 | $+3.33$ $[-5.74, +12.96]$ | $+4.44$ $[-3.05, +12.30]$ | $+8.70^{***}$ $[+3.15, +15.37]$ |
| `junk_fees` | R2 | 68.46 | 65.06 | 66.11 | 64.57 | $+3.40^{\dagger}$ $[-0.04, +6.70]$ | $+2.35^{\dagger}$ $[-0.39, +5.06]$ | $+3.89^{**}$ $[+1.14, +6.51]$ |
| `context_effects` | R4 | 76.04 | 71.98 | 70.94 | 76.94 | $+4.06$ $[-11.25, +21.32]$ | $+5.10$ $[-8.04, +16.52]$ | $-0.90$ $[-11.96, +8.71]$ |
| `preference_redistribution` | R1 | 80.07 | 77.13 | 78.57 | 78.79 | $+2.94^{***}$ $[+1.25, +4.64]$ | $+1.50$ $[-0.83, +3.81]$ | $+1.28$ $[-1.30, +3.84]$ |
| `accuracy_nudges` | R2 | 68.23 | 61.93 | 64.73 | 66.20 | $+6.30^{***}$ $[+2.33, +10.40]$ | $+3.50$ $[-2.59, +10.10]$ | $+2.03$ $[-3.67, +7.02]$ |
| `hiring_algorithms` | R0 | 72.66 | 72.96 | 73.91 | 74.55 | $-0.30$ $[-2.93, +2.17]$ | $-1.25$ $[-3.47, +1.02]$ | $-1.89^{\dagger}$ $[-4.22, +0.28]$ |
| `story_beliefs` | R3 | 65.35 | 64.79 | 65.68 | 67.49 | $+0.56$ $[-2.81, +3.78]$ | $-0.33$ $[-3.01, +2.29]$ | $-2.15$ $[-5.49, +1.22]$ |
| `infotainment` | R0 | 64.37 | 65.11 | 66.67 | 68.02 | $-0.75$ $[-3.25, +1.92]$ | $-2.30^{\dagger}$ $[-5.08, +0.40]$ | $-3.65^{*}$ $[-6.98, -0.48]$ |
| `affective_priming` | R1 | 76.53 | 76.59 | 75.28 | 75.23 | $-0.06$ $[-2.46, +2.63]$ | $+1.25$ $[-1.34, +4.26]$ | $+1.30$ $[-1.46, +4.42]$ |
| `quantitative_intuition` | R2 | 71.56 | 69.18 | 62.42 | 62.58 | $+2.38$ $[-5.82, +11.04]$ | $+9.14^{*}$ $[+0.99, +16.99]$ | $+8.99^{*}$ $[+0.03, +18.04]$ |
| `obedient_twins` | R0 | 80.83 | 80.28 | 79.45 | 80.14 | $+0.56$ $[-1.94, +3.06]$ | $+1.38$ $[-1.67, +4.83]$ | $+0.69$ $[-2.64, +4.03]$ |
| **Row mean (15 rows)** | | **70.91** | 67.54 | 67.24 | 68.06 | $\mathbf{+3.37}$ | $\mathbf{+3.67}$ | $\mathbf{+2.85}$ |

three iteration-eligible sub-studies are discussed in greater depth, chosen to span the per-sub-study lift distribution in Table 4. Each structure ends with a frozen behavioral-disposition summary and self-report reliability note (gated by the validator), not shown. The full selected-round extraction prompts are released under `iteration/final/final_extraction_<study>.md` (one file per sub-study, 19 in total; for the 6 zero-shot sub-studies the file is identical to `iteration/round0/round0_extraction_<study>.md`).

**Three observations from the table.** *(i) Structure size concentrates at* 4–5 *sub-profiles.* 9 of 19 structures have 4 sub-profiles; 8 have 5; 2 have 6 (`obedient_twins` R3, `quantitative_intuition` R3—both reached the 6-sub-profile validator cap during iteration). *(ii) BDE's literal labels never reappear.* The headers `bg`, `dp`, `ep` of the hand-crafted template do not surface as sub-profile names in any of the 19 structures; the reasoning-style and evaluation axes are instead authored as sub-study-specific constructs (e.g., *Cognitive Reflection* for `accuracy_nudges`, *Decision Style* for `context_effects`, *Thinking-Style* for `quantitative_intuition`, *Inertia & Closure* for `default_eric`). *(iii) Demographic anchoring appears in roughly half the structures, never as "`bg`".* It surfaces as *Demographic Background Profile* (`consumer_minimalism`, `context_effects`, `default_eric`, `privacy`), *Demographic Anchor Profile* (`preference_redistribution`, `promiscuous_donors`, `targeting_fairness`, `idea_evaluation`), *Demographic & Occupational Fit Profile* (`hiring_algorithms`), *Platform Demographics Profile* (`recommendation_algorithms`), or is folded into *Political Identity Profile* (`accuracy_nudges`, `infotainment`, `junk_fees`). The remaining structures encode identity through trait or ideology dimensions instead of a dedicated demographic section.

**Example 1 –** `consumer_minimalism` **($r^{\star}$=R4, sig win vs. both baselines; Table 4).** This is the pipeline's lone significant win against both baselines. Five sub-profiles: **Direct Preference**

Table 11: Top-level sub-profile names per Mega-Study sub-study at the iteration pipeline's selected round, in order of appearance in the extraction prompt. Iter-eligible rows show the calibration-50-selected round $r^\star$ (Table 4); zero-shot rows (italicised in the round column) use the round-0 LLM-generated prompt unchanged. **No sub-profile name appears in all 19 structures**; the closest recurring pattern is a demographic or political-identity anchor under varying names, present in roughly half the structures. The bg/dp/ep labels of the hand-crafted BDE template do not appear as headers in any of the 19 selected-round structures.

| Sub-study | $r^\star$ | Sub-profile names (in order) |
|---|---|---|
| accuracy_nudges | R4 | Political Identity Profile; Cognitive Reflection Profile; News Source Trust Profile; Sharing Norm Profile; Analytical Skepticism Profile |
| affective_priming | R0 | Affective Reactivity Profile; Self-Concept Anchor Profile; Values & Identity Anchor Profile; Self-Expression Profile; Empathic Resonance Profile |
| consumer_minimalism | R4 | Direct Preference Profile; Aesthetic Identity Profile; Demographic Background Profile; Social Evaluation Profile; Cross-Domain Consistency Profile |
| context_effects | R1 | Decision Style Profile; Risk and Loss Profile; Attribute Priority Profile; Baseline Choice Profile; Demographic Background Profile |
| digital_certification | R0 | Status & Display Profile; Luxury Familiarity Profile; Tech Adoption Profile; Self-Concept & Aesthetic Profile |
| hiring_algorithms | R0 | Demographic & Occupational Fit Profile; Tech Attitude Profile; Algorithm Trust Profile; Work Values Profile |
| infotainment | R0 | Political Identity Profile; Source Trust Profile; Social Conformity Profile; Entertainment Tolerance Profile |
| junk_fees | R0 | Political Identity Profile; Consumer Knowledge Profile; Fairness Norms Profile; Regulatory Focus & Trust Profile |
| obedient_twins | R3 | Attitude Stability Profile; Interpersonal Disposition Profile; Self-Efficacy Profile; Political Anchor Profile; Compliance & Reflection Profile; Epistemic Certainty & Help-Seeking Profile |
| preference_redistribution | R0 | Demographic Anchor Profile; Political Identity Profile; Communal vs. Agentic Values Profile; Economic-Anxiety & Lived-Experience Profile; Trust & Fairness-Norm Profile |
| privacy | R0 | Tech Habits Profile; Trust & Cynicism Profile; Autonomy & Control Profile; Demographic Background Profile |
| quantitative_intuition | R3 | Thinking-Style Profile; Behavioral Evidence Profile; Metacognitive Calibration Profile; Organizational Context Profile; Organizational QI Assessment Profile; Response-Style Profile |
| story_beliefs | R0 | Narrative Schema Profile; Affective Forecasting Profile; Cultural Background Profile; Genre Preference Profile; Engagement Calibration Profile |
| *Zero-shot sub-studies (no calibration items; round-0 prompt held fixed across all rounds)* | | |
| default_eric | *R0* | Inertia & Closure Profile; Pro-Social & Green Values Profile; Regulatory Focus Profile; Demographic Background Profile |
| idea_evaluation | *R0* | Creativity Sensitivity Profile; AI-Trust & Source-Salience Profile; Aesthetic Judgment Profile; Response-Style & Acquiescence Profile; Demographic Anchor Profile |
| idea_generation | *R0* | Cognitive Ability Profile; Divergent Thinking Profile; Domain Familiarity Profile; Self-Expression Profile |
| promiscuous_donors | *R0* | Demographic Anchor Profile; Political Identity Profile; Cynicism vs. Charity Profile; Fairness & Reciprocity Profile; Civic Engagement Profile |
| recommendation_algorithms | *R0* | Platform Demographics Profile; Digital Behavior Profile; Cognitive & Algorithmic Sophistication Profile; Autonomy & Self-Concept Profile |
| targeting_fairness | *R0* | Fairness & Communal Values Profile; Privacy & Tech Profile; Political & Identity Profile; Demographic Anchor Profile |

**Profile**, **Aesthetic Identity Profile**, **Demographic Background Profile**, **Social Evaluation Profile**, and **Cross-Domain Consistency Profile** (added during iteration). Iteration grew the structure from R0's 4 sub-profiles to R4's 5 by adding the *cross-domain consistency* axis, which the meta-extractor introduced after R3's diff-log diagnosed insufficient coverage of the voluntary-versus-forced minimalism disambiguator.

**Example 2 –** `accuracy_nudges` **($r^\star$=R4, not significant against either baseline; see Table 4).** Five sub-profiles, growing from R0's 4 by the addition of an **Analytical Skepticism Profile** at R4. The headline gain over raw is small and the contrast against structured is slightly negative; this is an example of a sub-study where iteration produces a directional improvement on raw but where the gain is within bootstrap noise at $n$=30.

**Example 3 –** `obedient_twins` **($r^\star$=R3, trending loss vs. raw but not significant; see Table 4).** The largest directional deficit in Table 4 on an iteration-eligible sub-study. Iteration grew the structure to 6 sub-profiles (**Attitude Stability**, **Interpersonal Disposition**, **Self-Efficacy**, **Political Anchor**, **Compliance & Reflection**, **Epistemic Certainty & Help-Seeking**), hitting the validator's sub-profile cap. The pattern resembles overfitting: calibration-50 picked R3, but the selected round's holdout accuracy on the 30 evaluation personas falls below raw, suggesting the calibration trajectory and the eval trajectory diverged for this sub-study. This is the failure mode that the information-flow protocol does not eliminate—calibration-50 is a leakage-clean signal, not a guarantee of holdout generalization.

# G Appendix: Twin-2K-500 cognitive-bias and pricing question examples

This appendix reproduces a small set of representative items from the Twin-2K-500 held-out evaluation block referenced in §3.1 and Appendix A. All wordings are reproduced verbatim from the prompt files the simulator received, located at `Digital-Twin-Simulation/text_simulation/text_questions/pid_*.txt`.

## G.1 Cognitive-bias items (3 of the 17 held-out heuristics-and-biases items)

**Asian-disease framing Tversky and Kahneman [1981] (gain frame).** *Imagine that the U.S. is preparing for the outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimate of the consequences of the programs are as follows: If Program A is adopted, 400 people will die. If Program B is adopted, there is 1/3 probability that nobody will die, and 2/3 probability that 600 people will die. Which of the two programs would you favor?* Six-point single choice from *I strongly favor program A* to *I strongly favor program B*.

**Linda's conjunction problem Tversky and Kahneman [1983].** *Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.* A six-point likelihood matrix (*Extremely improbable* → *Extremely probable*) is then elicited on three statements: (a) *Linda is a teacher in an elementary school*; (b) *Linda works in a bookstore and takes Yoga classes*; (c) *Linda is a bank teller and is active in the feminist movement*. The conjunction signature is the rate at which respondents rank (c) above (a).

**Mental-accounting jacket Tversky and Kahneman [1981].** *Imagine that you go to purchase a jacket for $250. The jacket salesperson informs you that the jacket you wish to buy is on sale for $240 at the other branch of the store which is ten minutes away by car. Would you drive to the other store?* Binary Yes / No.

## G.2 Pricing items (3 of the 40-item willingness-to-pay block)

Each pricing item presents one branded product with a posted price; the response is binary *Yes, I would purchase the product* / *No, I would not purchase the product*. Per-item dimensions vary along (i) product category, (ii) brand salience, and (iii) deviation of the posted price from typical shelf price.

**Tylenol Extra Strength (commodity, in-range price).** *Please consider the following product category: Pain Remedies – Headache. Suppose you are in a grocery store, and you see the following product in that category: Tylenol Extra Strength Caplets with 500 mg Acetaminophen, 100 Ct. The product is priced at: $2.19. Would you or would you not purchase this product?*

**Land O Lakes Salted Stick Butter (commodity, mid-range price).** *Please consider the following product category: Dairy Products. Suppose you are in a grocery store, and you see the following product in that category: Land O Lakes Salted Stick Butter, 16 oz, 4 Sticks. The product is priced at: $7.39. Would you or would you not purchase this product?*

**Goya Cooked Ham (deliberately above shelf price).** *Please consider the following product category: Refrigerated Deli Meats. Suppose you are in a grocery store, and you see the following product in that category: Goya Cooked Ham 16 oz. The product is priced at: $53.98. Would you or would you not purchase this product?* The posted price is several times typical shelf, so this item probes whether the simulator correctly rejects the offer.

# H Broader impacts

This work studies how to structure persona information for LLM-based digital twins. Potential benefits include more sample-efficient behavioral measurement, improved evaluation of persona-based simulators, and better tools for studying heterogeneity in decision-making without repeatedly collecting new responses. At the same time, more accurate persona simulation can create risks if used for privacy-invasive profiling, manipulative personalization, or targeting vulnerable individuals. We do not release respondent-level private data or a deployable model; the released artifacts are extraction prompts, pipeline code, and discovered structure templates. Future deployments should require consent for persona-data use, limit sensitive-attribute inference, and evaluate privacy and fairness risks before applying digital twins in consequential settings.

# I Compute resources.

All experiments are inference-only against hosted APIs, so the dominant cost is token usage. We summarize per-call and per-experiment costs under a single set of assumptions: a raw Twin-2K-500 transcript is approximately $32 \times 10^3$ input tokens, a structured persona is approximately $4 \times 10^3$ tokens, an extraction prompt is approximately $2 \times 10^3$ tokens, and a holdout item plus simulator response together account for approximately $0.7 \times 10^3$ tokens. Using public list prices for `gpt-5.4-nano` ($0.10/$0.40 per million input/output tokens), one extractor call costs approximately $0.005 per persona and one simulator call costs approximately $0.0006 with a structured persona or $0.0034 with the raw transcript. Aggregated to the experiments reported in the paper, a full Twin-2K-500 sweep over the BDE, unstructured, and raw conditions ($n$=50 personas, 88 holdout items) costs approximately $20 on `gpt-5.4-nano`; one round of the Mega-Study auto-discovery loop on a single sub-study costs approximately $0.50, so the full five-round loop across all 19 sub-studies costs approximately $50; deployment-time inference on the selected round costs approximately $10 per simulator model. Robustness reruns on `gpt-5.4-mini` and `Qwen3-8B` scale these totals by the corresponding per-token price ratios. End-to-end reproduction of all reported numbers is therefore on the order of a few hundred US dollars in API spend; the paired bootstrap with $B$=10,000 resamples and all parsing and scoring run on a single CPU and contribute negligibly to cost.

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